

App.jsx

```
import { useState, useEffect } from 'react';
import Header from './components/Header';
import Footer from './components/Footer';
import CourseCard from './components/CourseCard';
import StudentProfile from './components/StudentProfile';

function App() {

  // Task 3 Step 71–73: fetch from API, loading + error state
  const [courses, setCourses] = useState([]);
  const [loading, setLoading] = useState(true);
  const [error, setError] = useState(null);

  // Task 2 Step 68: search input state
  const [searchTerm, setSearchTerm] = useState("");

  // Task 2 Step 69: enrolled courses state (lifted up)
  const [enrolledCourses, setEnrolledCourses] = useState([]);

  // Task 3 Step 71: useEffect to fetch on mount ([ ] = runs once after mount)
  useEffect(() => {
    fetch('https://jsonplaceholder.typicode.com/posts')
      .then(res => res.json())
      .then(posts => {
        // map first 5 posts to course-like objects
        const mapped = posts.slice(0, 5).map((post, i) => ({
          id: post.id,
          code: `CS10${i + 1}`,
          name: post.title.slice(0, 30),
          credits: (i % 2 === 0) ? 4 : 3,
          grade: ['A', 'B', 'A', 'B', 'C'][i],
        }));
        setCourses(mapped);
        setLoading(false);
      })
      .catch(err => {
        setError(err.message);
        setLoading(false);
      });
  }, []);

  // Task 3 Step 75: log whenever courses state changes
```

```

// dependency array [courses] means this runs only when courses updates,
// not on every render — without it, this would run after every render
useEffect(() => {
  console.log('Courses updated:', courses.length);
}, [courses]);

// Task 2 Step 69: enroll handler — lifts state up from CourseCard
function handleEnroll(course) {
  const alreadyEnrolled = enrolledCourses.find(c => c.id === course.id);
  if (!alreadyEnrolled) {
    setEnrolledCourses([...enrolledCourses, course]);
  }
}

// Task 2 Step 68: filter courses by search term
const filteredCourses = courses.filter(c =>
  c.name.toLowerCase().includes(searchTerm.toLowerCase())
);

return (
  <>
    {/* Task 1 Step 64: siteName passed as prop, enrolledCount from state */}
    <Header siteName="EduCore Student Portal" enrolledCount={enrolledCourses.length}
  />

  <main style={{ maxWidth: '960px', margin: '24px auto', padding: '0 16px' }}>

    <h2 style={{ marginBottom: '12px' }}>Courses</h2>

    {/* Task 2 Step 68: search input */}
    <input
      type="text"
      placeholder="Search courses..."
      value={searchTerm}
      onChange={e => setSearchTerm(e.target.value)}
      style={{ padding: '8px', marginBottom: '16px', width: '100%' }}
    />

    {/* Task 3 Step 72: loading state */}
    {loading && <p>Loading...</p>}

    {/* Task 3 Step 73: error state */}
    {error && <p style={{ color: 'red' }}>Error: {error}</p>}
  </main>
)

```

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    { /* Task 2 Step 67: map over filtered courses */
    <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(200px,
1fr))', gap: '16px' }}>
      {filteredCourses.map(course => (
        <CourseCard
          key={course.id}
          {...course}
          onEnroll={() => handleEnroll(course)}
        />
      ))}
    </div>

    <StudentProfile />

  </main>

  <Footer />
</>
);
}

```

export default App;

Data.js

```

export const courses = [
  { id: 1, code: 'CS101', name: 'Intro to Programming', credits: 4, grade: 'A' },
  { id: 2, code: 'CS201', name: 'Data Structures', credits: 4, grade: 'B' },
  { id: 3, code: 'MA101', name: 'Calculus I', credits: 3, grade: 'A' },
  { id: 4, code: 'EC101', name: 'Circuit Theory', credits: 3, grade: 'B' },
  { id: 5, code: 'ME101', name: 'Engineering Mechanics', credits: 3, grade: 'C' },
];

```

Header.jsx

// Task 1 Step 62 & 64: Header component — receives siteName and enrolledCount as props

```

function Header({ siteName, enrolledCount }) {
  return (
    <header style={{ background: '#1e3a5f', color: 'white', padding: '16px 24px' }}>
      <h1>{siteName}</h1>
      <nav>
        <a href="#" style={{ color: 'white', marginRight: '16px' }}>Home</a>
        <a href="#" style={{ color: 'white', marginRight: '16px' }}>Courses</a>
        <a href="#" style={{ color: 'white' }}>Profile</a>
      </nav>
    </header>
  );
}

```

```

        </nav>
        { /* Task 2 Step 70: enrolled count passed as prop */ }
        <p>Enrolled: {enrolledCount}</p>
    </header>
  );
}

export default Header;

```

Footer.jsx

```

// Task 1 Step 63: Footer component
function Footer() {
  return (
    <footer style={{ background: '#1e3a5f', color: '#fff', padding: '16px', textAlign: 'center' }}>
      <p>&copy; 2026 EduCore College Management System</p>
    </footer>
  );
}

export default Footer;

```

Coursecard.jsx

```

// Task 1 Step 65: CourseCard — accepts name, code, credits, grade, onEnroll as props
function CourseCard({ name, code, credits, grade, onEnroll }) {
  return (
    <div style={{ border: '1px solid #ddd', borderRadius: '4px', padding: '16px', background:
'#fafafa' }}>
      <h3>{code}</h3>
      <p>{name}</p>
      <p>{credits} Credits | Grade: {grade}</p>
      { /* Task 2 Step 69: Enroll button — calls handler passed from App */ }
      <button onClick={onEnroll} style={{ marginTop: '8px', padding: '6px 12px', cursor:
'pointer' }}>
        Enroll
      </button>
    </div>
  );
}

export default CourseCard;

```

Studentprofile.jsx

```
import { useState } from 'react';

// Task 3 Step 74: StudentProfile — own local state, form with onChange handlers
function StudentProfile() {
  const [name, setName] = useState("");
  const [email, setEmail] = useState("");
  const [semester, setSemester] = useState("");

  return (
    <div style={{ border: '1px solid #ddd', borderRadius: '4px', padding: '20px', marginTop: '24px' }}>
      <h2>Student Profile</h2>
      <div style={{ marginTop: '12px' }}>
        <input
          type="text"
          placeholder="Name"
          value={name}
          onChange={e => setName(e.target.value)}
          style={{ display: 'block', marginBottom: '8px', padding: '8px', width: '100%' }}
        />
        <input
          type="email"
          placeholder="Email"
          value={email}
          onChange={e => setEmail(e.target.value)}
          style={{ display: 'block', marginBottom: '8px', padding: '8px', width: '100%' }}
        />
        <input
          type="text"
          placeholder="Semester"
          value={semester}
          onChange={e => setSemester(e.target.value)}
          style={{ display: 'block', padding: '8px', width: '100%' }}
        />
      </div>
      {name} && <p style={{ marginTop: '12px' }}>Hello, {name}! Semester: {semester}</p>
    </div>
  );
}

export default StudentProfile;
```

Output:



